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COMPUTEX and GTC Taipei preview: From GPU Racks to AI Factories

CITI'S TAKE

COMPUTEX 2026 is likely to mark a major shift in the AI infrastructure narrative. In the previous cycle, investor attention was concentrated on GPUs, HBM, CoWoS capacity, and Blackwell rack shipments. This year, the center of gravity is moving broader: NVIDIA is expected to emphasize not only Rubin GPUs, but also Vera CPUs, Spectrum-X networking, CMX context memory, CPO, liquid cooling, and high-voltage power architecture. The key message is that AI infrastructure is no longer just a GPU cluster. Other than Nvidia's supply chain, we expect to see more on CPU orchestration, Ethernet AI fabrics and CPO, liquid cooling and 800V DC power, BMC telemetry..etc. Computex is likely to become a full AI factory event, where CPU orchestration, GPU acceleration, memory hierarchy, networking bandwidth, cooling, and power delivery are jointly optimized.

Various rack configuration demonstration — Based on NVIDIA previous announcement during GTC in March, it appears to be defining several rack categories: GPU racks, Vera CPU racks, LPU racks, CMX context-memory racks, Spectrum networking racks, power racks, and cooling/liquid distribution infrastructure (see our report on [GTC implications](#) in Mar). This is no longer a conventional server deployment model. It is a composable AI factory model. We expect to see various different rack solutions from major ODMs including Hon Hai, Quanta, Wistron/Wiwynn, etc.

GPU rack remains the compute anchor. CPU and LPU rack to show up — Rubin NVL72 and Rubin Ultra NVL144/NVL576 extend the same logic following the GB system, but at far higher power density and network complexity. Vera Rubin Ultra NVL576 combines eight separate MGX NVL racks, each with 72 Rubin Ultra GPUs, into one 576-GPU NVLink domain using copper and direct optical connections. This suggests that the "rack" is becoming only one module inside a larger pod-scale compute domain. On CPU, we believe Vera CPU could become one of COMPUTEX 2026's key highlights, with expected Vera CPU production preparation of about 2m units in 2026. We expect Hon Hai and Quanta to demonstrate their Vera Rack/system during Computex.

CPO, Power and Cooling Are the Next Constraints — Spectrum-X and CPO highlight that data movement is becoming as important as compute. Optical engines, FAUs, silicon photonics, switches and fiber management have become more strategic. At the same time, rack power may rise from 40–60kW in Hopper to 180–220kW+ in Rubin, pushing 800V DC, power shelves and liquid cooling adoption. We expect Hon Hai and Quanta to demonstrate their SPX rack and system and Delta to demonstrate their power solution.

Positive on Taiwan technology supply chain — Beyond TSMC and CoWoS, the opportunity increasingly includes ABF substrates, sockets, thermal modules, power shelves, busbars, optical assembly, BMC controllers, and rack integration. We believe the most important COMPUTEX 2026 message is that AI infrastructure is entering into a new system-design phase.

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From GPU Racks to AI Factories

Computex 2026 will kick off on June 1st next week. The keynote lineup starting from Nvidia, Qualcomm, Marvell, Intel and NXP represent four different layers of the AI buildout: edge AI, connectivity, compute platforms and industrial/automotive intelligence. We see AI performance is increasingly constrained not only by accelerator supply, but also by networking, optical interconnect, custom silicon and system-level bandwidth. COMPUTEX should confirm that the next AI cycle is moving from “more GPUs” to “better infrastructure”. Meanwhile, NVIDIA’s separate GTC Taipei presence at COMPUTEX will likely remain the main sentiment driver, as Jensen Huang’s keynote is positioned around the future of AI and new technology. The key implication is that Taiwan’s supply chain will be judged less by single-component exposure and more by its ability to deliver complete AI systems: racks, power, thermal, networking, PCB/substrate, advanced packaging, testing and manufacturing scale.

Rack Configurations: From NVL72 to Modular AI Factory Blocks

The most important architectural transition is that NVIDIA is moving from server-level design to rack-level and eventually pod-level design. Blackwell GB200 NVL72 was the first major step in this direction, turning the rack itself into the system boundary. Rubin takes this much further. Based on NVIDIA previous announcement during GTC in March, it appears to be defining several rack categories: **GPU racks, Vera CPU racks, LPU racks, CMX context-memory racks, Spectrum networking racks, power racks, and cooling/liquid distribution infrastructure**. (see our report on [GTC implications](#) in Mar). This is no longer a conventional server deployment model. It is a composable AI factory model.

The GPU rack remains the compute anchor. Blackwell NVL72 combines 72 GPUs into a rack-scale NVLink domain, with direct liquid cooling and rack-level power delivery. Rubin NVL72 and Rubin Ultra NVL144/NVL576 extend the same logic, but at far higher power density and network complexity. **NVIDIA’s official developer blog states that Vera Rubin Ultra NVL576 combines eight separate MGX NVL racks, each with 72 Rubin Ultra GPUs, into one 576-GPU NVLink domain using copper and direct optical connections. This suggests that the “rack” is becoming only one module inside a larger pod-scale compute domain. We expect ODMs including Hon Hai, Quanta, Wistron/Wiwynn, Asustek, Inventec and Pegatron to demonstrate VR NVL72 during Computex.**

Figure 1. Key highlights of Nvidia's AI demonstration during Computex

Rack Type	Main Role	Key Components	Strategic Meaning
Rubin GPU rack	Main AI acceleration	GPUs, NVLink, HBM, liquid cooling	Defines rack-scale compute density
Vera CPU rack	Agentic AI orchestration	Vera CPUs, SOCAMM, NICs, BMC	Raises CPU content and memory bandwidth needs
Spectrum networking rack	Scale-out AI fabric	Spectrum-X / Spectrum-6 switches	Connects GPU, CPU, and CMX domains
Power rack	High-density power delivery	Power shelves, busbars, BBU	Supports 200kW–600kW rack class
Cooling system / CDU	Thermal infrastructure	CDU, manifolds, QDs, sensors	Enables full liquid AI factory deployment

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CPU Strategy: Vera Is Not a Traditional Server CPU

Vera CPU may be one of the most important COMPUTEX themes. NVIDIA's CPU strategy does not appear to simply take on Intel and AMD in the general-purpose server market. Instead, Vera is designed to control the AI factory orchestration layer. Vera positions as optimized for high single-threaded performance and energy efficiency to complement NVIDIA GPUs, rather than as a broad general CPU lineup.

In agentic AI, the CPU workload is not just feeding GPUs. It includes request scheduling, agent workflow control, tool-call coordination, memory indexing, retrieval management, simulation, networking control, and system telemetry. The CPU becomes the “traffic controller” for AI factories. As inference workloads become longer and more persistent, CPU-side memory bandwidth and capacity also matter more. It is also important to note that SOCAMM design is important for Vera as SOCAMM gives Vera a denser, lower-power, high-bandwidth memory architecture, likely better suited for AI orchestration than conventional DIMM-heavy server designs. And with AI inference becoming more agentic and CPU-intensive, CPU demand expands alongside GPU demand instead of being cannibalized by acceleration.

During Nvidia's latest earnings call, the CEO Jensen Huang reaffirmed NVIDIA's expectation for a roughly US\$200bn CPU market opportunity, with Vera CPU positioned as part of NVIDIA's broader AI infrastructure push. We believe Vera CPU could become one of COMPUTEX 2026's key highlights, with expected Vera CPU production preparation of about 2m units in 2026. **We expect Hon Hai and Quanta to demonstrate their Vera Rack/system during Computex.**

Networking and CPO: Spectrum-X Becomes the AI Fabric Layer

Networking will be another major COMPUTEX focus. As AI systems scale from single racks to pods, the bottleneck shifts from chip compute to data movement. We note that hyperscalers prefer Ethernet economics and operational familiarity, but AI requires much tighter congestion control, telemetry, routing, and workload-aware optimization. NVIDIA's Spectrum-X roadmap points toward Ethernet

becoming a specialized AI fabric rather than a generic datacenter network. **We would expect ODMs including Hon Hai/Foxconn and Quanta to demonstrate its SPX rack solution and switch.**

CPO will likely be one of the most debated COMPUTEX topics. NVIDIA has already positioned Spectrum-X Ethernet Photonics and Quantum-X Photonics as part of its AI factory networking roadmap. For Rubin Ultra, CPO becomes more important because the physical scale of NVL576 exceeds what copper can handle efficiently. **NVIDIA's developer blog supports the idea that NVL576 uses copper and direct optical connections across eight racks. Therefore, the practical adoption curve is likely: pluggable optics and LPO/NPO in 2025–2026, switch-side CPO and direct optical links in 2026–2027, and broader optical scale-up fabrics after Rubin Ultra.**

With more CPO penetration into the next 2 years, optical engines, silicon photonics PICs, FAUs, active alignment equipment, optical connectors, fiber management, and liquid-compatible photonic packaging all become more important. However, **we believe that FAU and OE should not be treated as mechanically always one-to-one.** One optical engine usually requires a fiber attach interface, but mapping depends on Tx/Rx layout, lane count, WDM architecture, and whether fibers are shared, split, or aggregated.

Cooling and Power: The AI Rack Becomes an Electrical-Thermal System

Cooling and power may become the most practical bottlenecks for Rubin deployment. Rubin NVL8 server references around 24kW TDP and over 32kW TDP at the 2U system level, while Rubin NVL72 rack estimates would be roughly 180kW rack TDP to 220kW rack TDP. Recall that Hopper racks could often operate around 40–60kW with air or hybrid cooling. Blackwell NVL72 pushes into 120–180kW, making direct liquid cooling standard. Rubin pushes rack power toward 200kW and beyond, while Rubin Ultra may move toward several hundred kilowatts.

Power delivery is also shifting. At 54V, a 216kW rack implies around 4,000A current, which becomes difficult in copper, busbars, connectors, and thermal management. This explains why NVIDIA will be pushing 800V DC, power racks, 110kW power shelves, BBU support, and DC datacenter architecture. **The future rack is not just a server cabinet. It is an electrical-thermal machine requiring coordinated power conversion, backup energy, liquid flow, telemetry, and safety systems.** We would expect to see more of the power and thermal solutions from Delta, AVC and Auras.

Figure 2. Power and cooling structure comparison for different generation

Generation	Approx. Rack Power Direction	Cooling Method	Power Architecture
Hopper	40–60kW	Air / partial liquid	Traditional AC PSU
Blackwell GB200	120–140kW	Direct liquid cooling	Rack power shelves / busbar emerging
GB300 / Blackwell Ultra	140–180kW	Full rack liquid cooling	Higher-density power shelves
Rubin NVL72	180–220kW+	Advanced rack liquid cooling	54V busbar near practical limits
Rubin Ultra	300–600kW potential	Pod/facility liquid loop	800V DC transition increasingly necessary

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